

Unit 1 :: Vector Calculus

☞ **PHY-02-MJ-MN-01-R-04: Mathematical Physics I** ☞

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Syllabus

Unit I: Vector calculus

- A. Review of vector algebra: addition and resolution of vectors; dot and cross products; scalar triple product; Cartesian components of a vector; conditions for perpendicularity and collinearity; scalar and vector fields.
- B. The del (∇) operator, gradient of scalar fields; normal and directional derivatives; divergence and curl of vector fields; geometrical interpretation of gradient, divergence and curl operations; the Laplacian (∇^2) operator; vector differential identities; the Jacobian.
- C. Notion of infinitesimal line, surface and volume elements; line, surface and volume integrals; flux of a vector field; Gauss' divergence theorem; Stokes' and Green's theorems (qualitative treatment only); applications of Gauss', Stokes' and Green's theorems.

1 Review of vector algebra

1.1 Addition of vectors

Two or more vectors may be added by geometrical laws, which are discussed below:

1. Triangle law of vector addition

Statement: If two vectors are represented both in magnitude and direction by two sides of a triangle taken in the *same order*, then their resultant vector is represented both in magnitude and direction by the third side of the triangle taken in the *opposite order*.

Note: Vectors are in the *same order* when one's origin coincides with the other's terminus; otherwise, they are in the *opposite order*.

2. Parallelogram law of vector addition

Statement: If two vectors acting at the same point are represented both in magnitude and direction by the adjacent sides of a parallelogram, then their resultant is represented both in magnitude and direction by the diagonal of that parallelogram passing through the same point.

3. Polygon law of vector addition

Statement: If several vectors are represented both in magnitude and direction by the sides of a polygon taken in order, then their resultant is represented both in magnitude and direction by the closing side taken in reverse order.

1.1.1 Properties of vector addition

i. Vector addition is *commutative*, i.e.

$$\vec{A} + \vec{B} = \vec{B} + \vec{A}$$

ii. Vector addition is *associative*, i.e.

$$\vec{A} + (\vec{B} + \vec{C}) = (\vec{A} + \vec{B}) + \vec{C}$$

iii. Vector addition is *distributive*, i.e.

$$\lambda(\vec{A} + \vec{B}) = \lambda\vec{A} + \lambda\vec{B}$$

where λ is a real number.

iv. The sum of two negative vectors is a null vector. So if vectors $\vec{A}$ and $\vec{B}$ are negatives of each other, i.e.

$$\vec{A} = -\vec{B}$$

then we have

$$\vec{A} + \vec{B} = \vec{0}$$

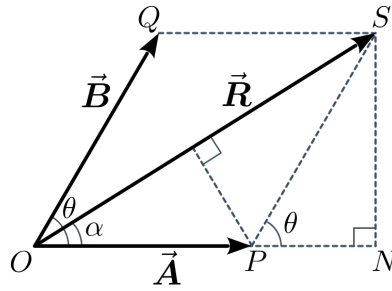


Figure 1: Magnitude of a vector sum

1.1.2 Magnitude of the vector sum

Given two vectors $\vec{A}$ and $\vec{B}$ and their resultant $\vec{R} = \vec{A} + \vec{B}$ as per the parallelogram law. Consider the construction as given in figure 1. Because now that $OPSQ$ is a parallelogram, we have

$$OP = QS = |\vec{A}| = A$$

$$OQ = SP = |\vec{B}| = B$$

$$\text{and } OS = |\vec{R}| = R$$

Let θ be the angle between vectors $\vec{A}$ and $\vec{B}$. Then we have $\angle SPN = \theta$. Hence, from $\triangle PSN$, we obtain

$$PN = SP \cos \theta = B \cos \theta$$

$$\text{and } SN = SP \sin \theta = B \sin \theta$$

Then in $\triangle OSN$, using Pythagoras' theorem, we get

$$\begin{aligned} OS^2 &= ON^2 + SN^2 \\ \Rightarrow OS^2 &= (OP + PN)^2 + SN^2 \\ \Rightarrow R^2 &= (A + B \cos \theta)^2 + (B \sin \theta)^2 \\ &= A^2 + B^2 \cos^2 \theta + 2AB \cos \theta + B^2 \sin^2 \theta \\ &= A^2 + B^2 (\cos^2 \theta + \sin^2 \theta) + 2AB \cos \theta \\ \Rightarrow R^2 &= A^2 + B^2 + 2AB \cos \theta \end{aligned}$$

Therefore,

$$|\vec{A} + \vec{B}| = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

1.1.3 Direction of the vector sum

Let α be the angle between the resultant vector $\vec{R} = \vec{A} + \vec{B}$. Because direction of the vector $\vec{A}$ is already known, knowing the angle α would enable us to know the direction of $\vec{R}$ with respect to $\vec{A}$.

From $\triangle OSN$, we obtain,

$$\begin{aligned}\tan \alpha &= \frac{SN}{ON} \\ &= \frac{SN}{OP + PN} \\ \Rightarrow \tan \alpha &= \frac{B \sin \theta}{A + B \cos \theta}\end{aligned}$$

Therefore,

$$\alpha = \tan^{-1} \left(\frac{B \sin \theta}{A + B \cos \theta} \right)$$

1.1.4 Subtraction on vectors

The subtraction of a vector from another may be defined in terms of the addition of one vector with the negative of the other vector, i.e.

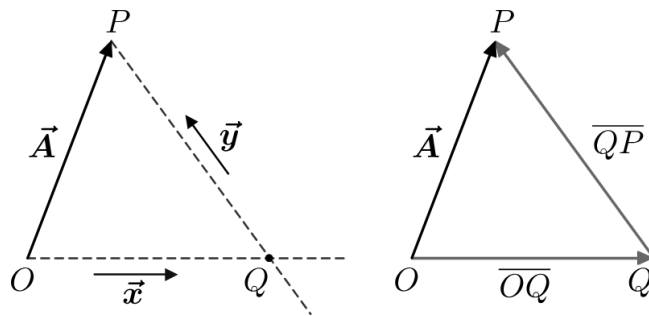
$$\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$$

The magnitude of the vector difference is:

$$|\vec{A} - \vec{B}| = \sqrt{A^2 + B^2 - 2AB \cos \theta}$$

1.1.5 Resolution of vectors

Suppose we have a certain vector $\vec{A}$ and two arbitrary vectors $\vec{x}$ and $\vec{y}$, all of them being coplanar. Let us now perform the following geometrical constructions.



- I. Through the origin O of vector $\vec{A}$, we draw a straight line parallel to $\vec{x}$.
- II. Through the terminus P of vector $\vec{A}$, we draw another straight line parallel to $\vec{y}$.
- III. Let the point of intersection of these two straight lines be Q .
- IV. We define the following two vectors:

$$\begin{aligned}\overline{OQ}: & \text{ origin at } O, \text{ terminus at } Q \\ \overline{QP}: & \text{ origin at } Q, \text{ terminus at } P\end{aligned}$$

Then, by triangle law of vector addition, we have

$$\vec{A} = \overrightarrow{OQ} + \overrightarrow{QP} \quad (1)$$

We know that when a vector is multiplied by some scalar, we obtain another parallel vector with different magnitude. Conversely, for two parallel vectors, one of them can be expressed as a scalar multiple of the other.

Consequently, because $\overrightarrow{OQ}$ is parallel to $\vec{x}$, we have

$$\overrightarrow{OQ} = \alpha \vec{x} \quad (2)$$

Also, because $\overrightarrow{QP}$ is parallel to $\vec{y}$, we have

$$\overrightarrow{QP} = \beta \vec{y} \quad (3)$$

where α, β are some arbitrary constant scalars.

Using equations (2) and (3) in equation (1) gives

$$\vec{A} = \alpha \vec{x} + \beta \vec{y} \quad (4)$$

Therefore, equation (4) shows that given some specific vector, it can be expressed as a vector sum of two arbitrary vectors, with each being multiplied by some constant scalars, and all of them lying on a plane.

Then we may say that the given vector has been resolved along two arbitrary vectors on the same plane, and the procedure followed to do so is known as the *resolution of a vector*.

1.2 Multiplication of vectors

1.2.1 Scalar product (or dot product)

Given two vectors $\vec{A}$ and $\vec{B}$ which have an angle θ between them, their scalar product is a scalar quantity which equals the product of their magnitudes and the cosine of the angle between them.

It is expressed mathematically as

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$$

or, $\boxed{\vec{A} \cdot \vec{B} = AB \cos \theta}$

where $A = |\vec{A}|$ and $B = |\vec{B}|$.

Some of the properties of the dot product are:

1. *Commutativity*: The order of vectors does not matter, i.e.

$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$$

2. *Distributivity*: The scalar product distributes over vector addition, i.e.

$$\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$$

3. The scalar product of a vector with itself equals the square of its magnitude, i.e.

$$\vec{A} \cdot \vec{A} = A^2$$

4. Multiplying either of the vectors by a scalar affects the scalar product by the same factor, i.e.

$$(\lambda \vec{A}) \cdot \vec{B} = \vec{A} \cdot (\lambda \vec{B}) = \lambda (\vec{A} \cdot \vec{B})$$

1.2.2 Vector product (or cross product)

Given two vectors $\vec{A}$ and $\vec{B}$ which have an angle θ between them, their vector product is another vector quantity which is perpendicular to the plane that contains both $\vec{A}$ and $\vec{B}$, and is denoted by $\vec{A} \times \vec{B}$.

- The magnitude of $\vec{A} \times \vec{B}$ is obtained as

$$|\vec{A} \times \vec{B}| = AB \sin \theta$$

- The direction of $\vec{A} \times \vec{B}$ is obtained using the right-hand thumb rule, which is applied as follows:

If the fingers of one's right hand are curled from the first vector to the second, then the stretched thumb of the right hand points in the direction of their cross product.

- The vector product is anti-commutative, i.e.

$$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$$

Because $\vec{A} \times \vec{B}$ and $\vec{B} \times \vec{A}$ are negatives of each other, the order of the cross product is important.

Some of the properties of the cross product are:

1. *Anti-commutativity*: The order of vectors is important, and reversing it changes the direction of the vector product, i.e.

$$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$$

2. *Perpendicularity*: The resultant vector product is perpendicular to the plane containing the original vectors.

3. *Magnitude*: The magnitude of the vector product is the area of the parallelogram formed by the two vectors.

4. *Distributivity*: The vector product distributes over vector addition, i.e.

$$\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$$

1.2.3 Scalar triple product

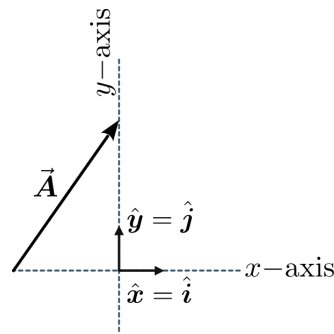
Given three vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$, their scalar triple product is given as

$$[\vec{A}\vec{B}\vec{C}] = \vec{A} \cdot (\vec{B} \times \vec{C})$$

It is calculated as the determinant of a 3×3 matrix formed by the component vectors, with a result of zero indicating the vectors are coplanar.

- The operation combines a dot product and a cross product acting on three vectors to produce a scalar quantity.

1.3 Cartesian components of a vector



Suppose we consider that, as a special case, $\hat{x}$ and $\hat{y}$ are the unit vectors which are perpendicular to each other, as we express

$$\hat{x} = \hat{i} \quad \text{and} \quad \hat{y} = \hat{j} \\ \therefore \hat{i} \perp \hat{j}$$

Also, we would have

$$|\hat{x}| = |\hat{i}| = 1 \\ \text{and } |\hat{y}| = |\hat{j}| = 1$$

We refer to these perpendicular directions as x -axis and y -axis.

Let us also express the constant scalars as

$$\alpha = A_x \quad \text{and} \quad \beta = A_y$$

Hence, from equation (4) we have

$$\vec{A} = \alpha \vec{x} + \beta \vec{y} \\ \Rightarrow \boxed{\vec{A} = A_x \vec{i} + A_y \vec{j}} \quad (5)$$

From equation (5), we can say that the vector $\vec{A}$ has been resolved into its Cartesian components A_x and A_y , which are given as

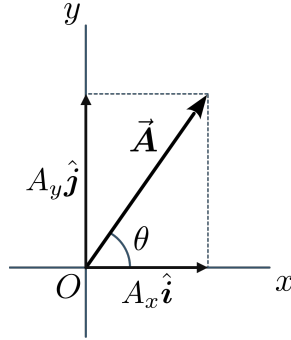
$$A_x = A \cos \theta \\ A_y = A \sin \theta$$

where θ is the angle between $\vec{A}$ and the x -axis.

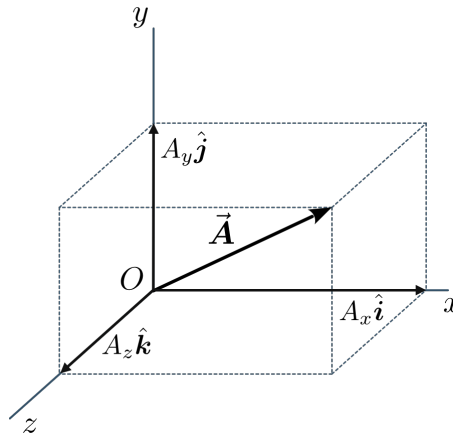
Conversely, we have,

$$A = \sqrt{A_x^2 + A_y^2}$$

$$\theta = \tan^{-1} \left(\frac{A_y}{A_x} \right)$$



We can extend the above idea to 3-dimensional space by including a third axis z which is perpendicular to both x and y axes.



If $\hat{k}$ is the unit vector along the z -axis, then we can resolve $\vec{A}$ into its three Cartesian components A_x , A_y and A_z as

$$\boxed{\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}}$$

Knowing the components A_x , A_y and A_z , the magnitude of $\vec{A}$ can be obtained as

$$\boxed{A = \sqrt{A_x^2 + A_y^2 + A_z^2}}$$

1.3.1 Test for perpendicular vectors

Given two vectors $\vec{A}$ and $\vec{B}$ which are perpendicular to one another, i.e the angle between them is $\theta = \frac{\pi}{2}$ rad.

Then their scalar product is

$$\vec{A} \cdot \vec{B} = AB \cos\left(\frac{\pi}{2}\right)$$

$$\Rightarrow \boxed{\vec{A} \cdot \vec{B} = 0}$$

Therefore, if $\vec{A} \perp \vec{B}$, then $\vec{A} \cdot \vec{B} = 0$ and vice versa.

1.3.2 Test for collinear vectors

When two vectors $\vec{A}$ and $\vec{B}$ are collinear, they may either be parallel or anti-parallel.

- a) If $\vec{A}$ is parallel to $\vec{B}$, then the angle between them is $\theta = 0$. Hence the magnitude of their vector product is

$$|\vec{A} \times \vec{B}| = AB \sin 0$$

$$\Rightarrow \boxed{|\vec{A} \times \vec{B}| = 0}$$

- b) If $\vec{A}$ is anti-parallel to $\vec{B}$, then the angle between them is $\theta = \pi$. Hence the magnitude of their vector product is

$$|\vec{A} \times \vec{B}| = AB \sin \pi$$

$$\Rightarrow \boxed{|\vec{A} \times \vec{B}| = 0}$$

Therefore, if $\vec{A} \parallel \vec{B}$, then $\vec{A} \times \vec{B} = \vec{0}$ and vice versa.

2 Vector differentiation

2.1 Scalar and vector fields

A field refers to a quantity whose value is a function of position in a region of space. There are two types of fields:

1. **Scalar field:** If for every point P at position $\mathbf{r} = (x, y, z)$ of a region R in space there corresponds a unique scalar ϕ , then $\phi(\mathbf{r})$ is known as a *scalar field*.

Examples:

- i. Temperature distribution in a rigid body
- ii. Local density of a liquid
- iii. Potential due to gravity

2. **Vector field:** If for every point P at position $\mathbf{r} = (x, y, z)$ of a region R in space there corresponds a unique vector $\mathbf{v}$, then $\mathbf{v}(\mathbf{r})$ is known as a *vector field*.

Examples:

- i. Heat flow rate in a conductor
- ii. Local velocity of a fluid
- iii. Electrostatic field

2.2 The vector differential operator (∇)

The vector differential operator, commonly denoted by the symbol ∇ and referred to as *del* or *nabla*, is a fundamental tool in vector calculus. In a three-dimensional Cartesian coordinate system, it is defined as:

$$\nabla \equiv \hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z}$$

2.3 Gradient of a scalar field

Given a scalar field $\phi(x, y, z)$, its gradient, denoted by $\nabla\phi$ or $\text{grad } \phi$, is given as

$$\nabla\phi = \text{grad } \phi = \hat{\mathbf{x}} \frac{\partial \phi}{\partial x} + \hat{\mathbf{y}} \frac{\partial \phi}{\partial y} + \hat{\mathbf{z}} \frac{\partial \phi}{\partial z}$$

Example: Find the gradient of the scalar field $\phi(x, y, z) = 2y^3 + 4xz + 3x$.

Solution: The partial derivatives of the given scalar field are:

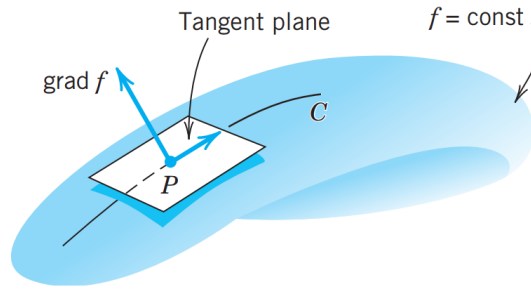
$$\begin{aligned} \frac{\partial \phi}{\partial x} &= \frac{\partial}{\partial x}(2y^3 + 4xz + 3x) & \frac{\partial \phi}{\partial y} &= \frac{\partial}{\partial y}(2y^3 + 4xz + 3x) & \frac{\partial \phi}{\partial z} &= \frac{\partial}{\partial z}(2y^3 + 4xz + 3x) \\ \implies \frac{\partial \phi}{\partial x} &= 4z + 3 & \implies \frac{\partial \phi}{\partial y} &= 6y^2 & \implies \frac{\partial \phi}{\partial z} &= 4x \end{aligned}$$

Therefore the gradient of ϕ is

$$\begin{aligned}\nabla\phi &= \hat{\mathbf{x}}\frac{\partial\phi}{\partial x} + \hat{\mathbf{y}}\frac{\partial\phi}{\partial y} + \hat{\mathbf{z}}\frac{\partial\phi}{\partial z} \\ \Rightarrow \nabla\phi &= \hat{\mathbf{x}}(4z+3) + \hat{\mathbf{y}}6y^2 + \hat{\mathbf{z}}4x\end{aligned}$$

2.3.1 Geometrical interpretation of gradient

Let S be a surface represented by $f(x, y, z) = c = \text{constant}$, where f is differentiable. Such a surface is called a *level surface* of f , and for different c we get different level surfaces. Now let C



be a curve on S through a point P of S . As a curve in space, C can be written as

$$\mathbf{r}(s) = \hat{\mathbf{x}}x(s) + \hat{\mathbf{y}}y(s) + \hat{\mathbf{z}}z(s)$$

We can thereby obtain a tangent vector $\mathbf{t}$ by differentiating $\mathbf{r}$ with respect to s , i.e.

$$\mathbf{t} = \frac{d\mathbf{r}}{ds} = \hat{\mathbf{x}}\frac{dx}{ds} + \hat{\mathbf{y}}\frac{dy}{ds} + \hat{\mathbf{z}}\frac{dz}{ds}$$

Because C lies on the surface S , we have

$$\begin{aligned}f[x(s), y(s), z(s)] &= c \\ \Rightarrow \frac{df}{ds} &= 0 \\ \Rightarrow \frac{\partial f}{\partial x}\frac{dx}{ds} + \frac{\partial f}{\partial y}\frac{dy}{ds} + \frac{\partial f}{\partial z}\frac{dz}{ds} &= 0 \\ \Rightarrow \underbrace{\left(\hat{\mathbf{x}}\frac{\partial f}{\partial x} + \hat{\mathbf{y}}\frac{\partial f}{\partial y} + \hat{\mathbf{z}}\frac{\partial f}{\partial z}\right)}_{=\nabla f} \cdot \underbrace{\left(\hat{\mathbf{x}}\frac{dx}{ds} + \hat{\mathbf{y}}\frac{dy}{ds} + \hat{\mathbf{z}}\frac{dz}{ds}\right)}_{=\mathbf{t}} &= 0 \\ \Rightarrow \nabla f \cdot \mathbf{t} &= 0 \\ \Rightarrow \nabla f \perp \mathbf{t}\end{aligned}$$

Therefore, the gradient of a scalar field at a point is normal (or perpendicular) to the tangent plane at that point.

2.3.2 Normal

If $\phi(x, y, z)$ is a scalar field, then we have

$$d\phi = \frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy + \frac{\partial\phi}{\partial z}dz \quad (1)$$

Now $\phi(x, y, z) = c$ represents a family of surfaces for different values of the constant c . Hence,

$$\begin{aligned} d\phi &= dc \\ \Rightarrow d\phi &= 0 \\ \Rightarrow \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz &= 0 \quad [\text{using eqn. (1)}] \end{aligned} \quad (2)$$

For a displacement $d\mathbf{r}$ along the surface, we have

$$d\mathbf{r} = \hat{\mathbf{x}} dx + \hat{\mathbf{y}} dy + \hat{\mathbf{z}} dz \quad (3)$$

We can now write equation (2) as

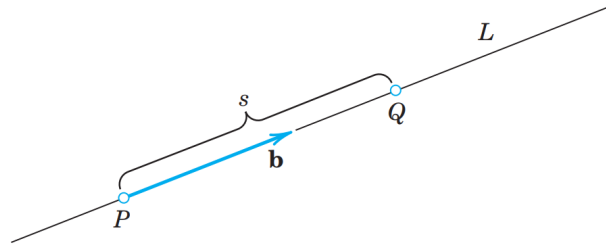
$$\begin{aligned} \left(\hat{\mathbf{x}} \frac{\partial \phi}{\partial x} + \hat{\mathbf{y}} \frac{\partial \phi}{\partial y} + \hat{\mathbf{z}} \frac{\partial \phi}{\partial z} \right) \cdot (\hat{\mathbf{x}} dx + \hat{\mathbf{y}} dy + \hat{\mathbf{z}} dz) &= 0 \\ \Rightarrow \nabla \phi \cdot d\mathbf{r} &= 0 \quad [\text{using eqn. (3)}] \\ \Rightarrow \nabla \phi &\perp d\mathbf{r} \end{aligned}$$

Because $d\mathbf{r}$ is directed along the surface and ∇ is perpendicular to it, hence the gradient ∇ is a vector which is normal to the surface $\phi(x, y, z) = c$.

2.3.3 Directional derivative

Given a scalar field $\phi(x, y, z)$, its directional derivative $D_{\mathbf{b}}\phi$ or $\frac{d\phi}{ds}$ at a point P in the direction of a unit vector $\mathbf{b}$ is defined by

$$D_{\mathbf{b}}\phi = \frac{d\phi}{ds} = \lim_{s \rightarrow 0} \frac{\phi(Q) - \phi(P)}{s} \quad (1)$$



As shown above, Q is a variable point on the straight line L in the direction of $\mathbf{b}$, and $|s|$ is the distance between P and Q .

$$\begin{aligned} s > 0 &\Rightarrow Q \text{ lies in the direction of } \mathbf{b} \text{ from } P \\ s < 0 &\Rightarrow Q \text{ lies in the direction of } -\mathbf{b} \text{ from } P \\ s = 0 &\Rightarrow Q = P \end{aligned}$$

Now, for any arbitrary point with position vector $\mathbf{r}$ that lies on the line L , its coordinates are all functions of s and we may write

$$\mathbf{r}(s) = \hat{\mathbf{x}}x(s) + \hat{\mathbf{y}}y(s) + \hat{\mathbf{z}}z(s) = \mathbf{p}_0 + s\mathbf{b}$$

where $\mathbf{p}_0$ is the constant position vector of the point P . Differentiating the above with respect to s , we obtain

$$\begin{aligned}\frac{d\mathbf{r}}{ds} &= \hat{\mathbf{x}} \frac{dx}{ds} + \hat{\mathbf{y}} \frac{dy}{ds} + \hat{\mathbf{z}} \frac{dz}{ds} = \underbrace{\frac{d\mathbf{p}_0}{ds}}_{=0} + \mathbf{b} \frac{ds}{ds} \\ \Rightarrow \hat{\mathbf{x}} \frac{dx}{ds} + \hat{\mathbf{y}} \frac{dy}{ds} + \hat{\mathbf{z}} \frac{dz}{ds} &= \mathbf{b}\end{aligned}\quad (2)$$

Because ϕ implicitly depends on s through x , y and z , we can use the chain rule to obtain

$$\begin{aligned}\frac{d\phi}{ds} &= \frac{\partial \phi}{\partial x} \frac{dx}{ds} + \frac{\partial \phi}{\partial y} \frac{dy}{ds} + \frac{\partial \phi}{\partial z} \frac{dz}{ds} \\ &= \left(\hat{\mathbf{x}} \frac{\partial \phi}{\partial x} + \hat{\mathbf{y}} \frac{\partial \phi}{\partial y} + \hat{\mathbf{z}} \frac{\partial \phi}{\partial z} \right) \cdot \left(\hat{\mathbf{x}} \frac{dx}{ds} + \hat{\mathbf{y}} \frac{dy}{ds} + \hat{\mathbf{z}} \frac{dz}{ds} \right) \\ \Rightarrow \frac{d\phi}{ds} &= \nabla \phi \cdot \mathbf{b} \quad [\text{using eqn. (2)}]\end{aligned}\quad (3)$$

From equations (1) and (3), we obtain

$$\boxed{D_{\mathbf{b}}\phi = \mathbf{b} \cdot \nabla \phi}$$

Example: Find the directional derivative of the scalar field $\phi(x, y, z) = 2x^2 + 3y^2 + z^2$ at point $P(2, 1, 3)$ in the direction of vector $\mathbf{a} = \hat{\mathbf{x}} - 2\hat{\mathbf{z}}$.

Solution: The gradient of the scalar field is

$$\nabla \phi = 4x\hat{\mathbf{x}} + 6y\hat{\mathbf{y}} + 2z\hat{\mathbf{z}}$$

The unit vector along the direction of the vector $\mathbf{a}$ is

$$\begin{aligned}\mathbf{b} &= \frac{\mathbf{a}}{|\mathbf{a}|} = \frac{\hat{\mathbf{x}} - 2\hat{\mathbf{z}}}{\sqrt{1^2 + (-2)^2}} \\ \Rightarrow \mathbf{b} &= \frac{1}{\sqrt{5}}(\hat{\mathbf{x}} - 2\hat{\mathbf{z}})\end{aligned}$$

The directional derivative of ϕ at any arbitrary point in the direction of $\mathbf{b}$ is

$$\begin{aligned}D_{\mathbf{b}}\phi &= \mathbf{b} \cdot \nabla \phi \\ &= \frac{1}{\sqrt{5}}(\hat{\mathbf{x}} - 2\hat{\mathbf{z}}) \cdot (4x\hat{\mathbf{x}} + 6y\hat{\mathbf{y}} + 2z\hat{\mathbf{z}}) \\ &= \frac{1}{\sqrt{5}}(4x - 4z) \\ \Rightarrow D_{\mathbf{b}}\phi &= \frac{4}{\sqrt{5}}(x - z)\end{aligned}$$

Therefore, at point $P(2, 1, 3)$, the directional derivative is

$$\begin{aligned}D_{\mathbf{b}}\phi(P) &= \frac{4}{\sqrt{5}}(2 - 3) \\ &= -\frac{4}{\sqrt{5}}\end{aligned}$$

2.4 Divergence of a vector field

Given a vector field $\mathbf{v}(x, y, z) = \hat{\mathbf{x}}v_x + \hat{\mathbf{y}}v_y + \hat{\mathbf{z}}v_z$, where v_x , v_y and v_z are the components of $\mathbf{v}$, its divergence, denoted by $\nabla \cdot \mathbf{v}$ or $\text{div } \mathbf{v}$, is given as

$$\nabla \cdot \mathbf{v} = \text{div } \mathbf{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$$

Example: Find the divergence of the vector field $\mathbf{v} = \hat{\mathbf{x}}3xz + \hat{\mathbf{y}}2xy - \hat{\mathbf{z}}yz^2$.

Solution: The components of the given vector field are

$$v_x = 3xz \quad v_y = 2xy \quad v_z = -yz^2$$

Hence, the required partial derivatives are

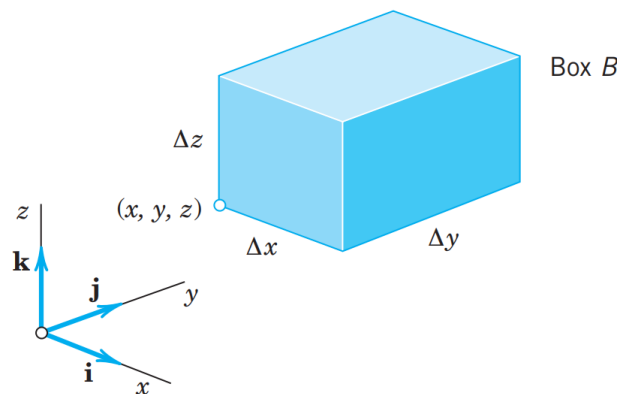
$$\begin{aligned} \frac{\partial v_x}{\partial x} &= \frac{\partial}{\partial x}(3xz) & \frac{\partial v_y}{\partial y} &= \frac{\partial}{\partial y}(2xy) & \frac{\partial v_z}{\partial z} &= \frac{\partial}{\partial z}(-yz^2) \\ \Rightarrow \frac{\partial v_x}{\partial x} &= 3z & \Rightarrow \frac{\partial v_y}{\partial y} &= 2x & \Rightarrow \frac{\partial v_z}{\partial z} &= -2yz \end{aligned}$$

Therefore, the divergence of $\mathbf{v}$ is

$$\begin{aligned} \nabla \cdot \mathbf{v} &= \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \\ \Rightarrow \nabla \cdot \mathbf{v} &= 3z + 2x - 2yz \end{aligned}$$

2.4.1 Geometrical interpretation of divergence

We consider the motion of a fluid in a region R having no sources or sinks in R , that is, no points at which fluid is produced or disappears. We assume that this fluid is compressible. Their density ρ depends on the coordinates x, y, z in space and may also depend on time t .



We consider the flow through a rectangular box B of small edges Δx , Δy , Δz parallel to the coordinate axes as shown in the figure above. The volume of the box is

$$\Delta V = \Delta x \Delta y \Delta z$$

Let $\mathbf{v} = \hat{\mathbf{x}}v_x + \hat{\mathbf{y}}v_y + \hat{\mathbf{z}}v_z$ be the velocity vector of the fluid. Then the vector field $\mathbf{u} = \rho\mathbf{v}$ gives the mass flow rate of the fluid (i.e. momentum per unit volume), written as

$$\begin{aligned}\mathbf{u} &= \rho(\hat{\mathbf{x}}v_x + \hat{\mathbf{y}}v_y + \hat{\mathbf{z}}v_z) \\ &= \hat{\mathbf{x}}u_x + \hat{\mathbf{y}}u_y + \hat{\mathbf{z}}u_z\end{aligned}$$

During a time interval Δt mass of fluid entering the face of cross-section area $\Delta x \Delta z$, at the face perpendicular to the y -axis as shown in the figure, is $(u_y)_y \Delta x \Delta z \Delta t = \frac{(u_y)_y \Delta V \Delta t}{\Delta y}$.

Likewise, the mass of fluid leaving the face of cross-section area $\Delta x \Delta z$ on the opposite face, perpendicular to the y -axis, is $(u_y)_{y+\Delta y} \Delta x \Delta z \Delta t = \frac{(u_y)_{y+\Delta y} \Delta V \Delta t}{\Delta y}$.

Hence, the mass lost across this pair of opposite faces perpendicular to the y -axis is

$$\begin{aligned}(\Delta m)_y &= \frac{(u_y)_{y+\Delta y} \Delta V \Delta t}{\Delta y} - \frac{(u_y)_y \Delta V \Delta t}{\Delta y} \\ &= \left[\frac{(u_y)_{y+\Delta y} - (u_y)_y}{\Delta y} \right] \Delta V \Delta t \\ \Rightarrow (\Delta m)_y &= \frac{\Delta u_y}{\Delta y} \Delta V \Delta t\end{aligned}$$

where $\Delta u_y = (u_y)_{y+\Delta y} - (u_y)_y$.

In a similar manner, the mass lost across the pairs of opposite faces perpendicular to the x and z axes can be shown to be

$$(\Delta m)_x = \frac{\Delta u_x}{\Delta x} \Delta V \Delta t \quad \text{and} \quad (\Delta m)_z = \frac{\Delta u_z}{\Delta z} \Delta V \Delta t$$

Therefore, the total mass lost during the time interval Δt is

$$\begin{aligned}\Delta m &= (\Delta m)_x + (\Delta m)_y + (\Delta m)_z \\ \Rightarrow \Delta m &= \left(\frac{\Delta u_x}{\Delta x} + \frac{\Delta u_y}{\Delta y} + \frac{\Delta u_z}{\Delta z} \right) \Delta V \Delta t \\ \Rightarrow \frac{\Delta m}{\Delta V \Delta t} &= \frac{\Delta u_x}{\Delta x} + \frac{\Delta u_y}{\Delta y} + \frac{\Delta u_z}{\Delta z}\end{aligned}$$

As the box becomes infinitesimally small, we have

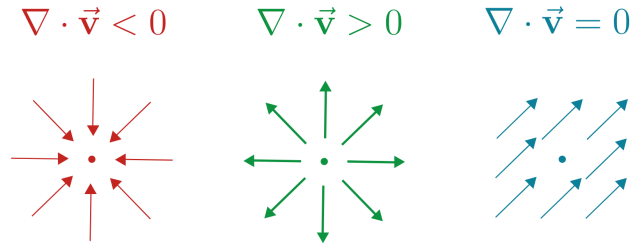
$$\Delta x, \Delta y, \Delta z, \Delta t, \Delta V, \Delta m \rightarrow 0$$

Thereby we get

$$\begin{aligned}\lim_{\Delta x, \Delta y, \Delta z \rightarrow 0} \left(\frac{\Delta u_x}{\Delta x} + \frac{\Delta u_y}{\Delta y} + \frac{\Delta u_z}{\Delta z} \right) &= \lim_{\Delta m, \Delta V \Delta t \rightarrow 0} \left(\frac{\Delta m}{\Delta V \Delta t} \right) \\ \Rightarrow \frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z} &= \frac{dm}{dV dt} \\ \Rightarrow \nabla \cdot \mathbf{u} &= \left\{ \begin{array}{l} \text{mass flow} \\ \text{per unit volume} \\ \text{per unit time} \end{array} \right\}\end{aligned}$$

Therefore, the divergence of a vector field at a point measures the net flow of the vector field per unit volume at the given point.

- $\nabla \cdot \mathbf{v} < 0$: The vector field $\mathbf{v}$ is *contracting inward*, i.e. acts as a *sink*.
- $\nabla \cdot \mathbf{v} > 0$: The vector field $\mathbf{v}$ is *expanding outward*, i.e. acts as a *source*.
- $\nabla \cdot \mathbf{v} = 0$: The vector field $\mathbf{v}$ is *incompressible*, i.e. amount entering a region equals the amount leaving. Such fields are said to be **solenoidal**.



2.5 Curl of a vector field

Given a vector field $\mathbf{v}(x, y, z) = \hat{\mathbf{x}}v_x + \hat{\mathbf{y}}v_y + \hat{\mathbf{z}}v_z$, where v_x , v_y and v_z are the components of $\mathbf{v}$, its curl, denoted by $\nabla \times \mathbf{v}$ or $\text{curl } \mathbf{v}$, is given as

$$\nabla \times \mathbf{v} = \text{curl } \mathbf{v} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ v_x & v_y & v_z \end{vmatrix}$$

$$\Rightarrow \nabla \times \mathbf{v} = \hat{\mathbf{x}} \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) + \hat{\mathbf{y}} \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) + \hat{\mathbf{z}} \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right)$$

Example: Find the curl of the vector field $\mathbf{v} = \hat{\mathbf{x}}yz + \hat{\mathbf{y}}3zx + \hat{\mathbf{z}}z$.

Solution: The components of the vector field are

$$v_x = yz \quad v_y = 3zx \quad v_z = z$$

Therefore the curl of $\mathbf{v}$ is

$$\begin{aligned} \nabla \times \mathbf{v} &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ v_x & v_y & v_z \end{vmatrix} \\ &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ yz & 3zx & z \end{vmatrix} \\ &= \hat{\mathbf{x}} \left[\frac{\partial}{\partial y}(z) - \frac{\partial}{\partial z}(3zx) \right] + \hat{\mathbf{y}} \left[\frac{\partial}{\partial z}(yz) - \frac{\partial}{\partial x}(z) \right] + \hat{\mathbf{z}} \left[\frac{\partial}{\partial x}(3zx) - \frac{\partial}{\partial y}(yz) \right] \\ &= \hat{\mathbf{x}}(0 - 3x) + \hat{\mathbf{y}}(y - 0) + \hat{\mathbf{z}}(3z - z) \\ \Rightarrow \nabla \times \mathbf{v} &= -\hat{\mathbf{x}}3x + \hat{\mathbf{y}}y + \hat{\mathbf{z}}2z \end{aligned}$$

2.6 The Laplacian (∇^2) operator

The Laplacian operator, denoted by ∇^2 , is a second order differential operator. In three-dimensional Cartesian coordinate system, it is defined as

$$\nabla^2 \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

In two- and one-dimensional Cartesian coordinates, the Laplacian operator is

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \quad \text{and} \quad \nabla^2 = \frac{d^2}{dx^2}$$

2.6.1 Laplacian of a vector field

Given a vector field $\mathbf{v} = \hat{\mathbf{x}}v_x + \hat{\mathbf{y}}v_y + \hat{\mathbf{z}}v_z$, its Laplacian is obtained by applying the ∇^2 operator on its scalar components, i.e.

$$\nabla^2 \mathbf{v} = \hat{\mathbf{x}}\nabla^2 v_x + \hat{\mathbf{y}}\nabla^2 v_y + \hat{\mathbf{z}}\nabla^2 v_z$$

Example: Find the Laplacian of $\phi = e^{xyz}$.

Solution: The partial derivatives of the given scalar field are:

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= yze^{xyz} \implies \frac{\partial^2 \phi}{\partial x^2} = y^2 z^2 e^{xyz} \\ \frac{\partial \phi}{\partial y} &= xze^{xyz} \implies \frac{\partial^2 \phi}{\partial y^2} = x^2 z^2 e^{xyz} \\ \frac{\partial \phi}{\partial z} &= xye^{xyz} \implies \frac{\partial^2 \phi}{\partial z^2} = x^2 y^2 e^{xyz} \end{aligned}$$

Therefore, the Laplacian of the given scalar field is

$$\begin{aligned} \nabla^2 \phi &= \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} \\ &= y^2 z^2 e^{xyz} + x^2 z^2 e^{xyz} + x^2 y^2 e^{xyz} \\ \implies \nabla^2 \phi &= (y^2 z^2 + x^2 z^2 + x^2 y^2) e^{xyz} \end{aligned}$$

2.7 Vector differential identities

2.7.1 Product rules

- (i) $\nabla(fg) = f\nabla g + g\nabla f$
- (ii) $\nabla(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A}$
- (iii) $\nabla \cdot (f\mathbf{A}) = f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f)$
- (iv) $\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$
- (v) $\nabla \times (f\mathbf{A}) = f(\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla f)$
- (vi) $\nabla \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A})$

2.7.2 Quotient rules

$$(i) \quad \nabla \left(\frac{f}{g} \right) = \frac{g \nabla f - f \nabla g}{g^2}$$

$$(ii) \quad \nabla \cdot \left(\frac{\mathbf{A}}{g} \right) = \frac{g(\nabla \cdot \mathbf{A}) - \mathbf{A} \cdot (\nabla g)}{g^2}$$

$$(iii) \quad \nabla \times \left(\frac{\mathbf{A}}{g} \right) = \frac{g(\nabla \times \mathbf{A}) + \mathbf{A} \times (\nabla g)}{g^2}$$

2.7.3 Second derivatives

$$(i) \quad \nabla \cdot (\nabla \phi) = \nabla^2 \phi$$

$$(ii) \quad \nabla \times (\nabla \phi) = \mathbf{0}$$

$$(iii) \quad \nabla \cdot (\nabla \times \mathbf{v}) = 0$$

$$(iv) \quad \nabla \times (\nabla \times \mathbf{v}) = \nabla(\nabla \cdot \mathbf{v}) - \nabla^2 \mathbf{v}$$

2.8 The Jacobian

In physical systems, coordinate transformations are often necessary to exploit the inherent symmetry of a problem, such as using polar coordinates for central force problems. When changing variables in multi-dimensional integration, the “volume” or “area” element changes. The Jacobian is the mathematical scaling factor that relates these differential elements between different coordinate systems.

Consider a transformation from variables (u, v) to (x, y) , where $x = x(u, v)$ and $y = y(u, v)$ are continuously differentiable functions. The Jacobian J of x and y with respect to u and v is defined by the following determinant:

$$J = \frac{\partial(x, y)}{\partial(u, v)} \equiv \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

For a three-dimensional system where x, y, z are functions of u, v, w , the above definition of the Jacobian can be generalized as the following 3×3 determinant:

$$J = \frac{\partial(x, y, z)}{\partial(u, v, w)} \equiv \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

The differential volume element transforms between the coordinate systems as

$$dx \, dy \, dz = \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du \, dv \, dw$$

Example: Find the differential volume element in spherical polar coordinates.

Solution: The transformation equations from spherical polar coordinates (r, θ, ϕ) to Cartesian coordinates (x, y, z) are as follows:

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

We obtain the Jacobian as

$$\begin{aligned} J = \frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} &= \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \phi} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \phi} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial \phi} \end{vmatrix} \\ &= \begin{vmatrix} \sin \theta \cos \phi & r \cos \theta \cos \phi & -r \sin \theta \sin \phi \\ \sin \theta \sin \phi & r \cos \theta \sin \phi & r \sin \theta \cos \phi \\ \cos \theta & -r \sin \theta & 0 \end{vmatrix} \\ &= \cos \theta (r^2 \sin \theta \cos \theta \cos^2 \phi + r^2 \sin \theta \cos \theta \sin^2 \phi) \\ &\quad + r \sin \theta (r \sin^2 \theta \cos^2 \phi + r \sin^2 \theta \sin^2 \phi) \\ &= r^2 \sin \theta \cos^2 \theta \underbrace{(\cos^2 \phi + \sin^2 \phi)}_{=1} + r^2 \sin^3 \theta \underbrace{(\cos^2 \phi + \sin^2 \phi)}_{=1} \\ &= r^2 \sin \theta \cos^2 \theta + r^2 \sin^3 \theta \\ &= r^2 \sin \theta \underbrace{(\cos^2 \theta + \sin^2 \theta)}_{=1} \\ \Rightarrow \left| \frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} \right| &= r^2 \sin \theta \end{aligned}$$

Therefore, the differential volume element in spherical polar coordinates is

$$\begin{aligned} dV &= dx \, dy \, dz \\ &= \left| \frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} \right| dr \, d\theta \, d\phi \\ \Rightarrow dV &= r^2 \sin \theta \, dr \, d\theta \, d\phi \end{aligned}$$

3 Vector integration

3.1 Infinitesimal elements

In vector calculus, infinitesimal elements are the fundamental building blocks used to perform vector integration over paths, surfaces and volumes. These elements represent tiny displacements, patches of area or chunks of volume that allow for the calculation of physical quantities such as work, flux and mass.

3.1.1 Line element

The infinitesimal line element (or displacement vector), often denoted as $d\mathbf{l}$ or $d\mathbf{r}$, represents an infinitesimal movement from one point to a nearby point. Its mathematical form depends on the coordinate system used:

- Cartesian coordinates: $d\mathbf{l} = dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}} + dz \hat{\mathbf{z}}$
- Spherical coordinates: $d\mathbf{l} = dr \hat{\mathbf{r}} + r d\theta \hat{\boldsymbol{\theta}} + r \sin \theta d\phi \hat{\boldsymbol{\phi}}$
- Cylindrical coordinates: $d\mathbf{l} = d\rho \hat{\boldsymbol{\rho}} + \rho d\phi \hat{\boldsymbol{\phi}} + dz \hat{\mathbf{z}}$

The line element is essential for line integrals, such as calculating the work done by a force $\mathbf{F}$ along a path C , given by $W = \int_C \mathbf{F} \cdot d\mathbf{l}$. It always points in the direction of increasing parameters along the path.

3.1.2 Surface element

The infinitesimal surface element, denoted as $d\mathbf{a}$ or $d\mathbf{S}$, is a vector quantity that represents a tiny patch of area. It has two defining characteristics: its magnitude is the infinitesimal scalar area dA , and its direction is normal (perpendicular) to the surface, represented by the unit vector $\hat{\mathbf{n}}$. Hence $d\mathbf{a} = \hat{\mathbf{n}} dA$.

The specific form of da depends on the orientation of the surface:

- For a surface of constant radius r in spherical coordinates (like a sphere):

$$d\mathbf{a} = r^2 \sin \theta d\theta d\phi \hat{\mathbf{r}}$$

- For a surface in the xy -plane:

$$d\mathbf{a} = \pm dx dy \hat{\mathbf{z}}$$

Surface elements are used to calculate the flux of a vector field through a surface, which is the integral $\int_S \mathbf{v} \cdot d\mathbf{s}$. For a closed surface, the normal vector $\hat{\mathbf{n}}$ is conventionally chosen to point outward.

3.1.3 Volume element

The infinitesimal volume element, denoted as dV or $d\tau$, represents a “tiny chunk” of space. Unlike line and surface elements, the volume element is a scalar. Its expression across different coordinate systems is:

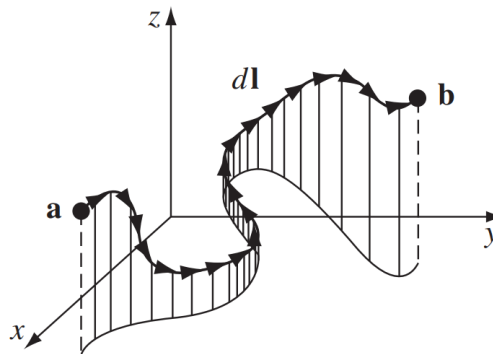
- Cartesian coordinates: $d\tau = dx dy dz$
- Spherical coordinates: $d\tau = r^2 \sin \theta dr d\theta d\phi$
- Cylindrical coordinates: $d\tau = \rho d\rho d\phi dz$

3.2 The line integral

A line integral is an expression of the form

$$\int_a^b \mathbf{v} \cdot d\mathbf{l}$$

where $\mathbf{v}$ is a vector field, $d\mathbf{l}$ is the infinitesimal displacement vector, and the integral is to be carried out along a prescribed path from point $\mathbf{a}$ to point $\mathbf{b}$, as shown in the figure below.



- If the path in question forms a closed loop C , then we denote the line integral of $\mathbf{v}$ over C as

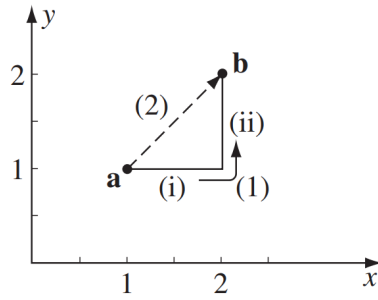
$$\oint_C \mathbf{v} \cdot d\mathbf{l}$$

- At each point on the path, we take the dot product of $\mathbf{v}$ (evaluated at that point) with the displacement $d\mathbf{l}$ to the next point along the path.
- The work done W by a force $\mathbf{F}$ in moving a particle along some path C is an example of a line integral, i.e.

$$W = \int_C \mathbf{F} \cdot d\mathbf{l}$$

- If the value of the line integral of a vector field $\mathbf{v}$ is independent of the path taken and depend only on the end points $\mathbf{a}$ and $\mathbf{b}$, then $\mathbf{v}$ is said to be a *conservative* vector field.

Example: Calculate the line integral of the vector field $\mathbf{v} = y^2\hat{\mathbf{x}} + 2x(y+1)\hat{\mathbf{y}}$ from the point $\mathbf{a} = (1, 1, 0)$ to the point $\mathbf{b} = (2, 2, 0)$ along the paths (1) and (2) as shown below.



Solution: The line element in Cartesian coordinates is

$$d\mathbf{l} = dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}} + dz \hat{\mathbf{z}}$$

- The path (1) can be considered to be split between two segments: a horizontal segment (i) and a vertical segment (ii). So we have

$$\int_{(1)} \mathbf{v} \cdot d\mathbf{l} = \int_{(i)} \mathbf{v} \cdot d\mathbf{l} + \int_{(ii)} \mathbf{v} \cdot d\mathbf{l}$$

- Along segment (i) $dy = dz = 0$ and $y = 1$, and so we have $d\mathbf{l} = dx \hat{\mathbf{x}}$. Hence

$$\begin{aligned} \mathbf{v} \cdot d\mathbf{l} &= [y^2\hat{\mathbf{x}} + 2x(y+1)\hat{\mathbf{y}}] \cdot (dx \hat{\mathbf{x}}) \\ \Rightarrow \mathbf{v} \cdot d\mathbf{l} &= \underbrace{y^2}_{y=1} dx = dx \end{aligned}$$

Hence,

$$\int_{(i)} \mathbf{v} \cdot d\mathbf{l} = \int_{x=1}^{x=2} \mathbf{v} \cdot d\mathbf{l} = \int_{x=1}^{x=2} dx = x \Big|_1^2 = 2 - 1 = 1$$

- Along segment (ii) $dx = dz = 0$ and $x = 2$, and so we have $d\mathbf{l} = dy \hat{\mathbf{y}}$. Hence

$$\begin{aligned} \mathbf{v} \cdot d\mathbf{l} &= [y^2\hat{\mathbf{x}} + 2x(y+1)\hat{\mathbf{y}}] \cdot (dy \hat{\mathbf{y}}) \\ \Rightarrow \mathbf{v} \cdot d\mathbf{l} &= \underbrace{2x(y+1)}_{x=2} dy = 4(y+1) dy \end{aligned}$$

Hence,

$$\int_{(ii)} \mathbf{v} \cdot d\mathbf{l} = \int_{y=1}^{y=2} \mathbf{v} \cdot d\mathbf{l} = \int_{y=1}^{y=2} 4(y+1) dy = 4 \left(\frac{y^2}{2} + y \right) \Big|_1^2 = 4 \left(\frac{2^2}{2} + 2 - \frac{1^2}{2} - 1 \right) = 10$$

Therefore, the line integral of the vector field $\mathbf{v}$ along the path (1) is

$$\begin{aligned} \int_{(1)} \mathbf{v} \cdot d\mathbf{l} &= \int_{(i)} \mathbf{v} \cdot d\mathbf{l} + \int_{(ii)} \mathbf{v} \cdot d\mathbf{l} \\ &= 1 + 10 = 11 \end{aligned}$$

- The path (2) can be described by the equation $y = x$. Then we have $dx = dy$ and also $dz = 0$. This gives us

$$\mathbf{v} = x^2 \hat{\mathbf{x}} + 2x(x+1) \hat{\mathbf{y}}$$

$$d\mathbf{l} = dx \hat{\mathbf{x}} + dx \hat{\mathbf{y}}$$

Hence

$$\begin{aligned} \mathbf{v} \cdot d\mathbf{l} &= [x^2 \hat{\mathbf{x}} + 2x(x+1) \hat{\mathbf{y}}] \cdot (dx \hat{\mathbf{x}} + dx \hat{\mathbf{y}}) \\ &= x^2 dx + 2x(x+1) dx \\ &= (3x^2 + 2x) dx \end{aligned}$$

Therefore, the line integral of the vector field $\mathbf{v}$ along the path (2) is

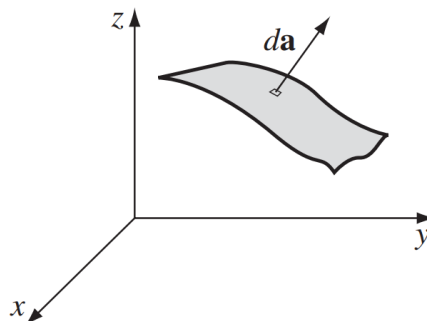
$$\begin{aligned} \int_{(2)} \mathbf{v} \cdot d\mathbf{l} &= \int_1^2 (3x^2 + 2x) dx \\ &= \left(3 \frac{x^3}{3} + 2 \frac{x^2}{2} \right)_1^2 \\ &= (x^3 + x^2)_1^2 = 2^3 + 2^2 - 1^3 - 1^2 \\ \Rightarrow \int_{(2)} \mathbf{v} \cdot d\mathbf{l} &= 10 \end{aligned}$$

3.3 The surface integral

A surface integral is an expression of the form

$$\int_S \mathbf{v} \cdot d\mathbf{a}$$

where $\mathbf{v}$ is some vector field, the integral is over a specified surface S , and $d\mathbf{a}$ is an infinitesimal patch of area, with direction perpendicular to the surface, as shown in the figure below.



- If the surface S in question is closed, then we denote the surface integral of $\mathbf{v}$ over S as

$$\oint_S \mathbf{v} \cdot d\mathbf{a}$$

- If $\mathbf{v}$ describes the flow of a fluid (mass per unit area per unit time), then $\int \mathbf{v} \cdot d\mathbf{a}$ represents the total mass per unit time passing through the surface. Hence the alternative name, “flux.”

3.4 The volume integral

A volume integral is an expression of the form

$$\int_V T d\tau$$

where T is a scalar field and $d\tau$ is an infinitesimal volume element.

- In Cartesian coordinates, the volume element is $d\tau = dx dy dz$, and so the volume integral is the following triple integral:

$$\int_V T d\tau = \iiint T dx dy dz$$

- The volume integral of a vector field $\mathbf{v} = v_x \hat{\mathbf{x}} + v_y \hat{\mathbf{y}} + v_z \hat{\mathbf{z}}$ is obtained by finding the volume integrals of its components separately, i.e.

$$\int_V \mathbf{v} d\tau = \hat{\mathbf{x}} \int_V v_x d\tau + \hat{\mathbf{y}} \int_V v_y d\tau + \hat{\mathbf{z}} \int_V v_z d\tau$$

3.5 Fundamental theorem for gradients

The line integral of the gradient of a scalar field equals the difference between the values of the field at the boundary points of the path of integration.

Given a scalar field T and a path bounded by the points $\mathbf{a}$ and $\mathbf{b}$, we have

$$\boxed{\int_{\mathbf{a}}^{\mathbf{b}} (\nabla T) \cdot d\mathbf{l} = T(\mathbf{b}) - T(\mathbf{a})}$$

Corollary 1

The integral $\int_{\mathbf{a}}^{\mathbf{b}} \nabla T \cdot d\mathbf{l}$ is independent of the path taken from $\mathbf{a}$ to $\mathbf{b}$.

Corollary 2

The integral $\oint \nabla T \cdot d\mathbf{l} = 0$ since the beginning and end points are identical, and hence $T(\mathbf{b}) - T(\mathbf{a}) = 0$.

3.6 Gauss' divergence theorem

The volume integral of the divergence of a vector field over a given volume equals the surface integral of the vector field over the closed surface bounding the volume.

Given a vector field $\mathbf{v}$ and a volume V which is bounded by the closed surface S , we have

$$\int_V (\nabla \cdot \mathbf{v}) d\tau = \oint_S \mathbf{v} \cdot d\mathbf{a}$$

3.7 Stokes' theorem

The surface integral of the curl of a vector field over a given open surface equals the line integral of the vector field over the closed curve bounding the surface.

Given a vector field $\mathbf{v}$ and a surface S which is bounded by the closed curve C , we have

$$\int_S (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \oint_C \mathbf{v} \cdot d\mathbf{l}$$

Corollary 1

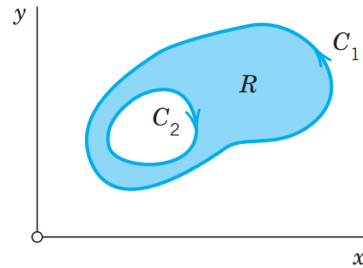
The integral $\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a}$ depends only on the boundary curve and not on the particular surface used.

Corollary 2

The integral $\oint (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = 0$ for *any closed surface*.

3.8 Green's theorem

Let R be a closed bounded region in the $x y$ -plane whose boundary C consists of finitely many smooth curves. Let $F_1(x, y)$ and $F_2(x, y)$ be functions that are continuous and have continuous partial derivatives $\partial F_1/\partial y$ and $\partial F_2/\partial x$ everywhere in some domain containing R . Then



Here we integrate along the entire boundary C of R in such a sense that R is on the left as we advance in the direction of integration, as shown in the figure above.